PROJECT DOCUMENTATION • OPERATIONAL PRESENTATION SCRIPT

JEEPTRACK: Smart PUJ Tracking & Fleet Dispatch Management System

System Architecture, Live Walkthrough Script, and Technical Glossary

Corridor: PLMun Main Campus ⇄ Alabang South Station

Document Code: JT-OPS-2026-v1.0

Target Date: Academic Year 2026–2027

Coverage: Manila South Road (National Highway)

Jurisdiction: Muntinlupa City, Philippines

Audience: Thesis Committee, Co-op Regulators, LGU

1.0 PRESENTATION SCHEDULE & AGENDA

Item	Agenda Topic	Key Objective	Allocated Time
1.1	Introduction & Problem Statement	Establish local transit delays, overcrowding, and lack of dispatch visibility.	2.0 Minutes
1.2	System Architecture	Overview of Commuter, Driver, and Central Dispatch operator roles.	2.5 Minutes
1.3	Live Walkthrough: Commuter Portal	Demonstrate real-time vehicle map, dynamic ETA countdown, and fare calculator.	3.5 Minutes
1.4	Live Walkthrough: Driver Console	Demonstrate driver navigation, passenger counters, and emergency SOS reporting.	3.5 Minutes
1.5	Live Walkthrough: Dispatch Admin	Demonstrate fleet KPIs, terminal geofence tracking, and passenger analytics.	3.5 Minutes
1.6	Technical Glossary & Panel Q&A	Address examiner questions and clarify underlying telemetry algorithms.	5.0 Minutes

2.0 BACKGROUND & PROBLEM STATEMENT

Public Utility Jeepneys (PUJs) remain the backbone of short-range public transport along the 5.2-kilometer corridor connecting the Pamantasan ng Lungsod ng Muntinlupa (PLMun) campus in Barangay Poblacion to the commercial and multimodal transport terminals in Alabang. Despite their critical importance, commuters, drivers, and transport operators face three persistent systemic challenges:

- **Unpredictable Commuter Waiting Times:** Students and local passengers have no mechanism to know when the next vehicle will arrive at designated stops, causing congestion at curb stops and excessive delays during morning and evening rush hours.
- **Unknown Seat Capacity:** Passengers frequently wait for approaching jeepneys only to discover upon arrival that the vehicle is completely full, leading to missed trips and dangerous roadside boarding attempts.

- **Lack of Real-Time Dispatch Oversight:** Transport cooperatives and local traffic managers operate without real-time tracking data, preventing automated terminal bay management, headway regulation (preventing vehicle bunching), and immediate response to vehicle breakdowns.

Project Objective: JEEPTRACK addresses these gaps by establishing a bidirectional, low-latency tracking and operations platform that synchronizes commuters, onboard drivers, and municipal dispatchers through standard web standards.

3.0 MASTER PRESENTATION SCRIPT (STEP-BY-STEP CUE SHEET)

The following table outlines the exact presenter script, corresponding visual cues, and key features to highlight during the evaluation.

SLIDE / STAGE & VISUAL CUE	SPOKEN NARRATION & TECHNICAL TALKING POINTS
Stage 1: Opening & Project Overview **Screen:** Landing Page / Title Slide showing system name, Muntinlupa corridor map, and KPI overview.	"Good day, members of the evaluation committee and guests. Today we present **JEEPTRACK**, a Smart Public Utility Jeepney Tracking and Fleet Dispatch Management System designed for the transit corridor between Pamantasan ng Lungsod ng Muntinlupa and Alabang. Our objective is straightforward: to bring real-time digital predictability to Philippine jeepney operations without requiring expensive proprietary hardware. The platform integrates three stakeholder interfaces: a Commuter Portal for passengers, an In-Cab Console for drivers, and a Central Dispatch Admin Portal for transport operators."
Stage 2: Commuter Portal Demonstration **Screen:** Navigate to Commuter View. Tap direction toggle (PLMun ➔ Alabang), click approaching PUJ marker, and inspect the slide-up card.	"Starting with the **Commuter Portal**: • **Interactive Corridor Map:** Commuters see live vehicle positions along Manila South Road. Vehicle icons dynamically rotate to reflect their actual heading compass angle. • **Dynamic ETA & Seat Availability:** Selecting an approaching vehicle reveals its unit code, speed in km/h, estimated arrival time in minutes, and real-time seating availability (for example, '14 / 20 Seats — 6 Free'). Commuters no longer have to guess if an oncoming jeepney can accept passengers. • **Automated Discount Fare Engine:** In compliance with LTFRB regulations, selecting any destination automatically calculates the official standard fare alongside the mandatory 20% discount for Students, PWDs, and Senior Citizens. • **Proximity Alert:** Commuters can activate a 2-minute arrival chime to receive notification when their selected unit approaches their boarding point."
Stage 3: Driver In-Cab Console Demonstration **Screen:** Switch to Driver View (`driver1`). Demonstrate Shift Toggle, Passenger '+' and '–' buttons, and Emergency SOS Help button.	"We now transition to the **Driver In-Cab Console**, designed specifically for in-vehicle smartphone mounts: • **Ergonomic Layout:** Large touch targets prevent driver distraction. At the top, the driver's assigned unit (PLM-01), name, license number, and a one-touch **ON DUTY / ON BREAK** status toggle are clearly visible. • **In-Cab Passenger Counter:** Drivers record passenger alighting and boarding with simple, thumb-sized '+' and '–' buttons, updating the commuter map in real time. • **One-Touch Emergency SOS:** In the event of a mechanical failure, road hazard, or medical emergency, pressing the Emergency SOS button immediately transmits coordinates to the control room and sounds an alarm siren. • **Road Hazard Reporting:** Drivers can report traffic congestion, lane closures, or localized flooding, warning both fellow drivers and commuters."

SLIDE / STAGE & VISUAL CUE	SPOKEN NARRATION & TECHNICAL TALKING POINTS
Stage 4: Central Dispatch Admin Portal **Screen:** Log in as Admin (`admin`). Show Dashboard KPIs, Terminal Bay Monitoring, and Operational Alerts Datatable.	"Finally, we demonstrate the **Central Dispatch Admin Portal**, which serves as the monitoring station for cooperative managers and local transport regulators: • **Operations KPI Overview:** Displays real-time counts for Active Units, Total Daily Passengers, On-Time Dispatch Reliability, and Active Safety Alerts. • **Automated Geofence Terminal Tracking:** Digital 80-meter perimeters around PLMun Terminal and Alabang South Station automatically log vehicle arrival, dock duration, and departure without manual paper clipboards. • **Incident Management:** When an emergency alert is received, dispatchers can inspect the vehicle's location, acknowledge the ticket, and coordinate response personnel. • **Integrated Enterprise Datatables:** Administrative records (Users, Fleet, Alerts, Terminals) include integrated search filters, CSV export capabilities, and pagination compliant with standard administrative dashboards."

4.0 TECHNICAL GLOSSARY & PLAIN-ENGLISH EXPLANATIONS

This section provides concise, formal definitions for technical terms utilized within the JEEPTRACK system architecture.

Technical Term	Formal Definition & Mechanism	Operational Purpose in JEEPTRACK
GPS Telemetry	The continuous electronic transmission of geospatial data packets comprising latitude, longitude, instantaneous speed, and bearing angle from onboard sensors to server endpoints.	Enables real-time rendering of moving vehicle markers and trajectory calculation on the corridor map.
Geofencing	A location-based technology that defines a virtual spatial boundary using geographic coordinates and a predetermined radial distance (e.g., 80 meters).	Automatically registers vehicle arrival, bay occupancy, and departure at PLMun and Alabang terminals without manual clerical input.
Dynamic ETA (Estimated Time of Arrival)	An algorithmic arrival calculation based on remaining route distance divided by real-time corridor velocity, updated dynamically as road conditions vary.	Provides commuters with realistic arrival countdowns that account for localized traffic delays.
Corridor Path Interpolation	Mathematical projection of coordinates along a predefined polyline trajectory corresponding to the Manila South Road transit geometry.	Ensures accurate visual mapping and provides continuous tracking simulation during academic evaluation and testing.
Headway Management	The temporal or spatial interval between successive public transit vehicles traversing the same route and direction.	Assists dispatchers in preventing 'vehicle bunching' (multiple empty units arriving simultaneously followed by long gaps).
Role-Based Access Control (RBAC)	An information security mechanism restricting system features and database operations based on verified user authorization (Commuter, Driver, Operator).	Protects fare matrices, dispatch controls, and administrative logs from unauthorized access or alteration.

5.0 COMMITTEE EVALUATION & DEFENSE Q&A

Q1: How does the system handle temporary cellular data loss or connectivity dropouts along the corridor?

> **Answer:** The system maintains state resilience by recording the last confirmed GPS timestamp and location. If a vehicle misses consecutive heartbeat pings, its status transitions to 'GPS Standby / Offline' with an explicit indicator (e.g., 'Last seen 3 minutes ago'). When mobile data reconnects, the telemetry pipeline re-establishes synchronization automatically. Central dispatch is immediately alerted if an active vehicle unexpectedly stops transmitting along an active route.

Q2: Can the platform support traditional Public Utility Jeepneys that lack built-in electronic control units?

> **Answer:** Yes. JEEPTRACK was intentionally engineered to avoid reliance on expensive proprietary hardware. Traditional PUJ operators only require an entry-level smartphone or tablet mounted on the dashboard running the driver web application. Modern Euro-4 and Electric PUJs can utilize the exact same interface or connect via onboard GPS telemetry hardware.

Q3: How are student and senior citizen discount fares verified to prevent fraudulent fare calculations?

> **Answer:** The Commuter Portal features an accredited digital transit profile displaying the verified student number, institutional identification, and photo badge. Commuters present this digital credential upon boarding, allowing drivers to confirm eligibility for the automated 20% legal discount computed by the fare matrix.

Q4: What was the architectural rationale for developing a Progressive Web Application (PWA) rather than native mobile applications?

> **Answer:** A high-performance web architecture eliminates client-side installation friction, app store approval delays, and smartphone storage constraints for students and drivers. The platform runs across Android, iOS, Windows, and macOS devices through any standard compliant browser while preserving high-performance vector mapping and responsive touch interaction.

6.0 FORMAL CLOSING STATEMENT

> "In conclusion, **JEEPTRACK** demonstrates how standard modern web technologies can resolve long-standing operational inefficiencies in Philippine public transit. By providing predictability to commuters, ergonomics to drivers, and data-driven visibility to transport cooperatives and local authorities, the system offers an accessible, practical blueprint for transit modernization in Muntinlupa City.
>
> Thank you for your time and evaluation. We welcome your questions, observations, and recommendations."
